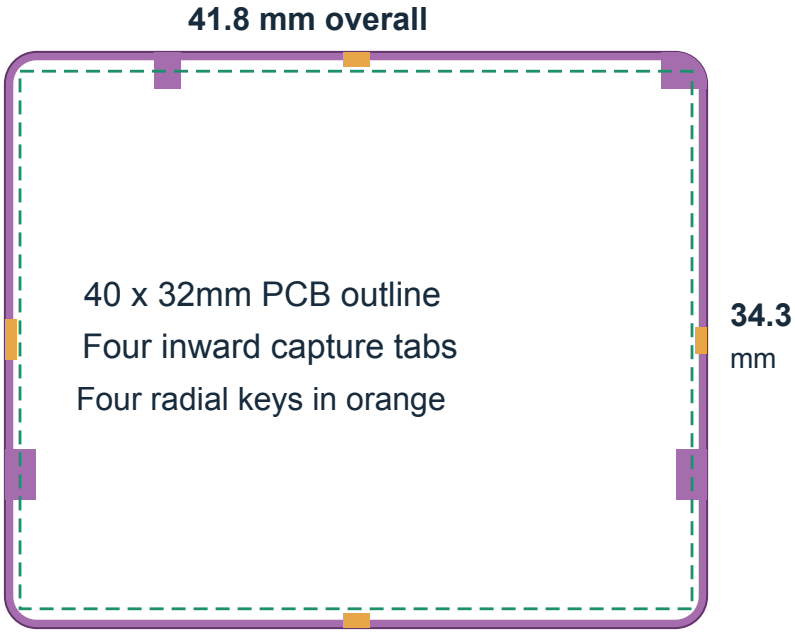
CT-RET-001 | Captive PCB frame / r11

PROTOTYPE — conditional thermal geometry only; physical retention, materials and ingress unqualified

Retainer plan



Frame thickness: 1.2mm; aperture: 40.8 x 33.3mm.

Board and its all-green Gerber remain unchanged.

Rigid enclosure: 57.4 x 49.4 x 21.0mm, plus docking key.

Assembly datums and checks

PCB support	Z=10.40mm; thickness 0.71–0.91mm
Guide / frame bottom	Z=11.50mm nominal
Lid stop plane	Z=13.00mm nominal
Maximum PCB top	Z=11.97mm in this model
SIM access	Open guide lane; no bridge above card
Component sweep	85 envelopes; minimum gap 0.205mm
Cable endpoint poses	27 cases; minimum body gap 0.235mm
Tab overlap	Guaranteed interior: ≥0.162mm² per tab
Thermal closure	≥0.058mm at assigned limits
Free play	Captive frame; zero movement is not claimed
Material	TECANAT PC candidate; not qualified

Factory closure and service

Insert SIM with power off and the board seated in the lifted carrier.

Fit the keys, inspect tabs/cables and reject any interference.

Thermal screen: -10 to +45 C, independent part temperatures and assigned CTE/tolerance bounds; not material guarantees.

Four radial slots allow expansion; assigned motion remains subject to tilt, load, stiffness, wear and physical testing.

Cable samples vary initial plug azimuth. Connector force/orientation, continuous movement and flex life are unqualified.

Contact stem is 7.50mm for the wider cavity; seal land unchanged. Pack/harness/dock and firmware remain unfinished.

No supplier message, order or physical water/dust qualification is implied by this drawing.